Group 14

Resource Coordination Platform for Community Resilience
Software Requirements Specification
For Subsystem

Version 1.0

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Software Requirements Specification

1. Introduction

1.1 Purpose

The Resource Coordination Platform for Community Resilience is a digital platform developed to improve the coordination of resources, volunteers, and relief activities during emergencies and community crises such as floods and landslides. It provides a centralized platform where community members can request assistance, donors can offer resources, volunteers can participate in relief activities, and coordinators can manage these activities through a unified dashboard. The system includes a responsive web application and a cross-platform mobile application to support efficient, transparent, and reliable emergency response management.

1.2 Scope

The Resource Coordination Platform is a cloud-based web and mobile application that supports disaster response and community resilience through organized resource management.

The platform enables:

- User registration and authentication
- Organization management
- Disaster resource requests
- Donation management
- Volunteer registration and task assignment
- Resource inventory management
- Real-time notifications
- Progress tracking
- Administrative monitoring

The system follows a Microservice Architecture, with each major functionality implemented as an independent service.

1.3 Definitions, Acronyms, and Abbreviations

Table 1.1: Definitions, Acronyms, and Abbreviations

Abbreviation	Definition
API	Application Programming Interface
CSV	Comma-Separated Values
Docker	Containerization platform used to package and deploy applications
HTTPS	Hypertext Transfer Protocol Secure
IAM	Identity and Access Management
JSON	JavaScript Object Notation
JWT	JSON Web Token
Kafka	Apache Kafka – Distributed event streaming platform
Kong	API Gateway responsible for routing and API management
PDF	Portable Document Format
PostgreSQL	Open-source relational database management system
RBAC	Role-Based Access Control
REST	Representational State Transfer
SRS	Software Requirements Specification
SSO	Single Sign-On
Swagger	OpenAPI-based API documentation tool
UUID	Universally Unique Identifier

1.4 References

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1.5 Overview

The remainder of this document is organized as follows.

- Chapter 1: Introduces the purpose, scope, terminology, references and overall structure of the document.
- Chapter 2: Provides a overview of the system including its product perspective main functions, user characteristics, operating environment, design constraints, assumptions, dependencies and requirement subsets
- Chapter 3: Describes the functional and non-functional requirements of the system in detail. Functional requirements are organized according to the platform's microservices, while non-functional requirements describe usability, reliability, performance, security, supportability, and design constraints.
- Chapter 4: Provides supporting information and references related to the requirements described in this document.

2. Overall Description

2.1 Product Perspective

ResQHub is a standalone software product developed to improve the coordination of community disaster response activities. Instead of relying on manual or paper-based processes, the platform brings emergency requests, volunteer coordination, resource management, and organizational collaboration together in one system.

The platform consists of a responsive web application for coordinators and administrators, a cross-platform mobile application for volunteers and community members, and a backend made up of independent microservices deployed using Docker containers.

2.2 Product Functions

Key functions of the system include:

- User registration, authentication and profile management.
- Organization registration and membership management.
- Disaster resource request submission and tracking
- Resource donation and inventory management
- Volunteer registration, approval and availability management.
- Disaster task creation and volunteer assignment.
- Resource allocation and inventory tracking.
- Real-time notifications and alerts.
- Administrative reporting and monitoring
- Audit logging and activity tracking.

2.3 User Characteristics

The system supports multiple categories of users.

Community Members	● Register ● Submit disaster assistance requests ● Monitor request progress ● Receive notifications regarding their requests
Donors	● Contribute disaster relief resources ● Monitor donation status ● Receive acknowledgement from participating organizations
Volunteers	● Register their skills ● Update their availability ● Receive assignments ● Report task progress ● Participate in relief operations
Organization Coordinators	● Manage assistance requests ● Assign tasks to volunteers ● Manage resources and inventories, within their organization.
System Administrators	● Manage organizations ● Assign system roles ● Verify organizations ● Monitor system activities and maintain overall platform integrity

2.4 Constraints

- Technical Constraints
  - Backend: Node.js + Express.js microservices
  - Frontend: React.js (web) and Flutter (mobile)
  - Database: PostgreSQL
  - Communication between microservices shall occur through REST APIs and Apache Kafka
  - Docker shall be used for containerization
  - System shall support real-time updates using WebSocket-based communication
  - The platform shall use a centralized API gateway implemented using Kong to manage API routing.
  - Cloud hosting: AWS/ Railway

- Operational Constraints

  - The system shall support operation during emergency situations where a large number of requests may be submitted within a short period.
  - User interface shall be simple and easy to use
  - The platform shall operate over internet-based networks
  - Organization coordinators shall be responsible for verifying requests, managing inventory, approving resource offers and assigning volunteers.
- Legal/ Social Constraints
  - The platform shall protect user's personal information and prevent unauthorized access to sensitive data
  - Access to user information shall be controlled using RBAC restrictions.
  - Data belonging to one organization shall not be accessible to unauthorized users of other organizations.

2.5 Assumptions and Dependencies

The following assumptions have been made during system development:

- Users have reliable internet connection when accessing the platform.
- Participating organizations are responsible for maintaining accurate operational information.
- Cloud infrastructure and database service remain available throughout system operation.
- The IAM framework, Kong API Gateway, PostgreSQL database, Kafka and Docker infrastructure function as expected.

2.6 Requirements Subset

The system requirements are divided into the following functional subsets to organize the development and implementation of the Resource Coordination Platform.

2.6.1 User Management Subset

This subset covers common functions available to registered users and organization staff.

- User registration and authentication
- User profile management
- Role and access management
- Organization and organization-member management
- Notifications
- User activity tracking

2.6.2 Help Request and Resource Donation Subset

This subset covers the main resource coordination activities performed through the mobile application.

- Submit and manage help requests
- Track help request status
- Offer resources and submit donations
- View and manage submitted donations
- Organization verification of help requests and donations
- Donation status tracking
- Resource allocation and fulfillment

2.6.3 Volunteer and Task Management Subset

This subset covers volunteer registration and emergency task coordination.

- Volunteer registration
- Volunteer profile and skill management
- Volunteer availability management
- Volunteer approval by organization administrators
- Creation and assignment of volunteer tasks
- Viewing assigned tasks
- Accepting and updating task progress

- Task completion and status tracking

2.6.4 Organization and Resource Management Subset

This subset covers administrative functions provided through the web application

- Organization management
- Help request verification and approval
- Donation verification
- Inventory management
- Resource allocation and restocking
- Volunteer management
- Task assignment and monitoring
- Dashboard and operational reporting
- Audit and activity monitoring

3. Specific Requirements

This section describes the functional and non-functional requirements of the Resource Coordination Platform for Community Resilience. The system uses a microservice architecture with seven independent services that support disaster response, resource coordination, volunteer management, and organizational collaboration. The following subsections describe the required functionalities and quality attributes in detail to support system design, implementation, and testing.

3.1 Functionality

The Resource Coordination Platform for Community Resilience uses a microservice-based architecture to support disaster resource management in a scalable, secure, and maintainable way. The platform consists of seven independent microservices: User Service, Organization Service, Resource Service, Request Service, Volunteer Service, Task Service, and Notification Service. Each service exposes RESTful APIs through the API Gateway and communicates with other services using synchronous HTTP requests and asynchronous Apache Kafka events where appropriate.
The platform supports different categories of users, including Community Members, Donors, Volunteers, Organization Coordinators, and System Administrators. Access to system functionality is controlled through Role-Based Access Control (RBAC) implemented by the selected IAM framework.

The following subsections describe the functional requirements of each microservice.

3.1.1 User Service

Description
The User Service manages user registration, authentication, authorization, profile management, and role management. It acts as the identity provider for the platform and integrates with the selected IAM framework to provide centralized authentication and authorization using JSON Web Tokens (JWTs).
Functional Requirements

Requirement ID	**FR-US-001**
Requirement Name	User Registration
Description	The system shall allow new users to register by providing valid personal information.
Actors	Community Member, Volunteer, Donor
Preconditions	The user does not already have an account.
Postconditions	A new user account shall be created and stored in the database.

Requirement ID	**FR-US-002**
Requirement Name	User Authentication
Description	The system shall authenticate registered users using the IAM framework. It shall validate credentials, generate JWT access and refresh tokens, attach user roles, reject invalid login attempts, and redirect users to their role-based dashboards.
Actors	Registered Users
Preconditions	User has a registered account.
Postconditions	User is successfully authenticated and granted access according to their role.

Requirement ID	**FR-US-003**
Requirement Name	Profile Management
Description	Authenticated users shall be able to view and update their personal information.
Actors	All Authenticated Users
Preconditions	User is authenticated.
Inputs	Full name, Phone number, Address, Profile picture, Notification preferences
Process	1. User views profile.2. User edits required fields.3. System validates modified information.4. Database is updated.
Postconditions	Updated profile information is stored successfully.

Requirement ID	**FR-US-004**
Requirement Name	Password Management
Description	The system shall provide functionality for changing and recovering passwords.
Actors	Registered Users
Preconditions	User is authenticated (for password change) or has a registered email (for password recovery).
Inputs	Current password (for change), New password, Recovery email/token
Process	1. User requests password change or recovery.2. System validates request. 3. Password policy is enforced.4. Password is updated.
Postconditions	User password is successfully changed or reset.

Requirement ID	**FR-US-005**
Requirement Name	Role Management
Description	System administrators shall assign or modify user roles.
Actors	System Administrator
Preconditions	Administrator is authenticated with sufficient privileges.
Inputs	User ID, New Role
Process	1. Administrator selects a user.2. Assigns or updates the user's role.3. System validates permissions.4. Updated role is saved and takes effect immediately.
Postconditions	User permissions are updated according to the assigned role.

Requirement ID	**FR-US-006**
Requirement Name	Account Status Management
Description	The system shall maintain and manage user account statuses.
Actors	System Administrator
Preconditions	User account exists.
Inputs	User ID, Account Status (Active, Pending Verification, Suspended, Disabled)
Process	1. Administrator updates account status.2. System stores the new status.3. Suspended or disabled users are denied access to protected resources.
Postconditions	Account status is updated and access permissions are enforced accordingly.

3.1.2 Organization Service

Description

The Organization Service manages organizations involved in disaster response activities. It allows organizations to register, manage members, define operational regions, and coordinate relief activities.

Functional Requirements

Requirement ID	**FR-ORG-001**
Requirement Name	Organization Registration
Description	Authorized users shall register organizations by providing the organization name, registration number, organization category, contact information, physical address, operating districts, and organization description. The system shall verify that duplicate organizations are not registered.
Actors	Authorized User
Preconditions	User is authenticated and authorized to register an organization.
Postconditions	Organization is successfully registered and stored in the database if no duplicate exists.

Requirement ID	**FR-ORG-002**
Requirement Name	Organization Verification
Description	Only System Administrators shall verify newly registered organizations. Verification statuses include Pending, Approved, and Rejected. Approved organizations shall be permitted to participate in disaster response operations.
Actors	System Administrator
Preconditions	Organization is registered with a Pending verification status.
Postconditions	Organization verification status is updated, and approved organizations gain access to disaster response operations.

Requirement ID	**FR-ORG-003**
Requirement Name	Organization Profile Management
Description	Organization Coordinators shall update organization details including contact information, office location, description, operating districts, and organization logo. The system shall maintain a version history of all profile modifications.
Actors	Organization Coordinator
Preconditions	Coordinator is authenticated and belongs to the organization.
Postconditions	Updated organization profile is stored and modification history is maintained.

Requirement ID	**FR-ORG-004**
Requirement Name	Organization Membership Management
Description	Organization Coordinators shall invite users to join their organizations. The invitation workflow includes sending invitations, accepting invitations, declining invitations, and removing members. The system shall maintain membership records for each organization.
Actors	Organization Coordinator
Preconditions	Organization exists and coordinator is authorized to manage members.
Postconditions	Membership records are updated based on invitation or membership actions.

Requirement ID	**FR-ORG-005**
Requirement Name	Organization Search
Description	Users shall search organizations using organization name, district, category, or verification status. Search results shall be paginated and sortable.
Actors	All Users

Preconditions	Organizations are available in the system.
Postconditions	Matching organizations are displayed according to the selected search criteria.

Requirement ID	**FR-ORG-006**
Requirement Name	Organization Reports
Description	Organization Coordinators shall generate reports summarizing the number of members, active volunteers, resource inventory, disaster requests handled, and task completion statistics. Reports shall be exportable in PDF and CSV formats.
Actors	Organization Coordinator
Preconditions	Organization data is available and the coordinator is authenticated.
Postconditions	Report is generated successfully and can be exported in PDF or CSV format.

3.1.3 Resource Service

Description

The Resource Service manages resource inventories, donations, allocation, and resource tracking. It provides up-to-date information about available disaster relief resources.

Functional Requirements

Requirement ID	**FR-RES-001**
Requirement Name	Resource Category Management
Description	Administrators shall create and maintain resource categories. Each category shall include a name, description, unit of measurement, and priority classification. Example categories include Food, Drinking Water, Medical Supplies, Clothing, Shelter Equipment, and Hygiene Kits.
Actors	System Administrator
Preconditions	Administrator is authenticated and authorized to manage resource categories.
Postconditions	Resource category is created or updated successfully in the system.

Requirement ID	**FR-RES-002**
Requirement Name	Donation Registration
Description	Donors shall register donated resources by specifying the resource category, quantity, unit, expiry date (if applicable), pickup location, and additional remarks. The system shall validate donation information before acceptance.
Actors	Donor
Preconditions	Donor is authenticated.
Postconditions	Donation is successfully registered with a pending verification status.

Requirement ID	**FR-RES-003**
Requirement Name	Donation Verification
Description	Organization Coordinators shall verify donated resources before adding them to inventory. Verification states include Pending, Accepted, and Rejected. Rejected donations shall include reasons for rejection.
Actors	Organization Coordinator
Preconditions	Donation has been registered and is awaiting verification.
Postconditions	Donation status is updated and accepted donations are added to inventory.

Requirement ID	**FR-RES-004**
Requirement Name	Inventory Management
Description	The system shall maintain inventory information including available quantity, reserved quantity, allocated quantity, remaining stock, warehouse location, and expiry information. Inventory shall automatically update whenever resources are allocated or consumed.

Actors	Organization Coordinator, System
Preconditions	Inventory records exist.
Postconditions	Inventory information is accurately updated and maintained.

Requirement ID	**FR-RES-005**
Requirement Name	Resource Allocation
Description	Authorized coordinators shall allocate available resources to approved requests. The system shall verify stock availability, reserve required quantities, update inventory records, record allocation history, and generate notification events through Apache Kafka.
Actors	Organization Coordinator
Preconditions	Approved request exists and sufficient inventory is available.
Postconditions	Resources are allocated, inventory is updated, and allocation history is recorded.

Requirement ID	**FR-RES-006**
Requirement Name	Inventory Search
Description	Users shall search inventory using resource category, availability, organization, expiry date, and storage location. The search interface shall support sorting and filtering.
Actors	All Users
Preconditions	Inventory data is available.
Postconditions	Matching inventory records are displayed according to the selected search criteria.

Requirement ID	**FR-RES-007**
Requirement Name	Inventory Alerts
Description	The system shall automatically generate alerts when inventory falls below minimum thresholds, resources approach expiry dates, or high-priority resources become unavailable. Alerts shall be forwarded to the Notification Service through Apache Kafka.
Actors	System
Preconditions	Inventory monitoring is active.
Postconditions	Alert events are generated and forwarded to the Notification Service.

Requirement ID	**FR-RES-008**
Requirement Name	Resource Reports
Description	The system shall generate analytical reports including current inventory levels, donation summaries, resource allocation statistics, resource utilization trends, and expired resource reports. Reports shall support export in PDF and CSV formats.
Actors	Organization Coordinator, System Administrator
Preconditions	Resource and inventory data are available.
Postconditions	Analytical report is generated successfully and can be exported in PDF or CSV format.

3.1.4 Request Service

Description

The Request Service manages disaster-related resource requests submitted by community members and organizations. It supports the full request lifecycle, including creation, review, approval, fulfillment, cancellation, and tracking. The service communicates with the Resource Service for inventory allocation, the Notification Service for event notifications, and the Task Service when volunteer assistance is required.

Functional Requirements

Requirement ID	**FR-REQ-001**
Requirement Name	Create Resource Request
Description	The system shall allow authenticated Community Members and Organization Coordinators to submit disaster relief resource requests. Each request shall include the resource category, resource type, required quantity, priority level, request location, required date, and emergency description.
Actors	Community Member, Organization Coordinator
Preconditions	User is authenticated and the user's account is active.
Postconditions	The request is stored in the database with a Pending status.

Requirement ID	**FR-REQ-002**
Requirement Name	View Requests
Description	Users shall view requests according to their authorization level. Community Members can view only their own requests, Organization Coordinators can view requests within their organizations, and System Administrators can view all requests. The request list shall support search, sorting, pagination, and filtering by status, priority, and location.
Actors	Community Member, Organization Coordinator, System Administrator
Preconditions	User is authenticated.
Postconditions	Authorized request records are displayed based on the user's permissions.

Requirement ID	**FR-REQ-003**
Requirement Name	Update Request
Description	Users shall update request details only while the request remains in the Pending state. Editable fields include quantity, description, priority, required date, and supporting documents. All modifications shall be recorded in an audit log.
Actors	Request Owner
Preconditions	User is authenticated and the request status is Pending.
Postconditions	Request details are updated and the modification is recorded in the audit log.

Requirement ID	**FR-REQ-004**
Requirement Name	Cancel Request
Description	Request owners shall cancel requests before approval. Upon cancellation, the request status becomes Cancelled, reserved resources (if any) are released, and a notification event is generated. Cancelled requests shall not proceed further.
Actors	Request Owner
Preconditions	User is authenticated and the request has not been approved.
Postconditions	Request status is updated to Cancelled, reserved resources are released, and a notification is generated.

Requirement ID	**FR-REQ-005**
Requirement Name	Approve Request
Description	Organization Coordinators shall review pending requests by validating request information and verifying resource availability before approving them. Approved requests shall transition to the Approved status.
Actors	Organization Coordinator
Preconditions	Request is in the Pending state.
Postconditions	Request status is updated to Approved.

Requirement ID	**FR-REQ-006**
Requirement Name	Reject Request
Description	Organization Coordinators shall reject requests when necessary. A rejection reason shall be provided. Rejected requests become read-only, and the requester shall receive a notification.
Actors	Organization Coordinator
Preconditions	Request is under review.
Postconditions	Request status is updated to Rejected, the rejection reason is recorded, and the requester is notified.

Requirement ID	**FR-REQ-007**
Requirement Name	Resource Allocation
Description	Approved requests shall be forwarded to the Resource Service, which shall reserve inventory, allocate resources, and update stock levels. If inventory is insufficient, the request shall remain partially fulfilled or pending according to organizational policy.
Actors	Resource Service, Organization Coordinator
Preconditions	Request has been approved.
Postconditions	Resources are allocated or the request status is updated according to resource availability.

Requirement ID	**FR-REQ-008**
Requirement Name	Fulfill Request
Description	When all requested resources have been delivered, the request status shall become Fulfilled. The fulfillment date shall be recorded, inventory updated, and a notification sent to the requester.
Actors	Organization Coordinator, Resource Service
Preconditions	All allocated resources have been delivered.
Postconditions	Request status is updated to Fulfilled, fulfillment information is recorded, inventory is updated, and notification is sent.

Requirement ID	**FR-REQ-009**
Requirement Name	Request Tracking
Description	Users shall monitor request progress through the statuses: Pending, Under Review, Approved, Rejected, Partially Fulfilled, Fulfilled, and Cancelled. Progress updates shall be displayed in chronological order.
Actors	Community Member, Organization Coordinator, System Administrator
Preconditions	Request exists in the system.
Postconditions	Users can view the current status and history of the request.

Requirement ID	**FR-REQ-010**
Requirement Name	Request History
Description	The system shall preserve historical request information, including request details, approval records, allocation records, status changes, and completion information. Historical records shall remain available for reporting purposes.
Actors	Organization Coordinator, System Administrator
Preconditions	Request lifecycle information is available.
Postconditions	Historical request records are stored and available for reporting and auditing.

3.1.5 Volunteer Service

Description
The Volunteer Service manages volunteer registration, approval, skills, availability, and task assignments.
Functional Requirements

Requirement ID	**FR-VOL-001**
Requirement Name	Volunteer Registration
Description	Authenticated users shall register as volunteers by providing their skills, certifications, experience, preferred working locations, and emergency contact details.
Actors	Authenticated User
Preconditions	User is authenticated and has an active account.
Postconditions	Volunteer registration is stored with a Pending verification status.

Requirement ID	**FR-VOL-002**
Requirement Name	Volunteer Verification
Description	Organization Coordinators shall review volunteer registrations. Verification statuses include **Pending**, **Approved**, and **Rejected**. Approved volunteers become eligible for task assignments.
Actors	Organization Coordinator
Preconditions	Volunteer registration exists and is awaiting verification.
Postconditions	Volunteer verification status is updated accordingly.

Requirement ID	**FR-VOL-003**
Requirement Name	Volunteer Profile Management
Description	Volunteers shall maintain their personal information, skills, certifications, availability, and contact information.
Actors	Volunteer
Preconditions	Volunteer is authenticated.
Postconditions	Volunteer profile information is updated successfully.

Requirement ID	**FR-VOL-004**
Requirement Name	Availability Management
Description	Volunteers shall update their availability status to **Available**, **Busy**, or **Unavailable**. Only volunteers marked as **Available** shall receive new task assignments.
Actors	Volunteer
Preconditions	Volunteer is authenticated.
Postconditions	Availability status is updated and reflected in task assignment decisions.

Requirement ID	FR-VOL-005
Requirement Name	Skill Management
Description	Volunteers shall add, modify, or remove skills. Example skills include Medical Assistance, Logistics, Transportation, Food Distribution, and Emergency Rescue. Skills shall be searchable.
Actors	Volunteer
Preconditions	Volunteer is authenticated.
Postconditions	Volunteer skills are updated and available for search.

Requirement ID	**FR-VOL-006**
Requirement Name	Volunteer Search

Description	Organization Coordinators shall search volunteers using skills, location, availability, organization, and experience. Search results shall support sorting and filtering.
Actors	Organization Coordinator
Preconditions	Volunteer records exist in the system.
Postconditions	Matching volunteer records are displayed according to the selected search criteria.

Requirement ID	**FR-VOL-007**
Requirement Name	Assignment History
Description	The system shall maintain volunteer assignment history, including assigned tasks, completion records, and performance ratings.
Actors	Volunteer, Organization Coordinator
Preconditions	Volunteer has participated in one or more task assignments.
Postconditions	Assignment history is stored and available for viewing and reporting.

Requirement ID	**FR-VOL-008**
Requirement Name	Volunteer Performance
Description	The system shall calculate volunteer performance statistics, including tasks completed, active assignments, average completion time, and participation frequency.
Actors	Organization Coordinator, System Administrator
Preconditions	Volunteer assignment data is available.
Postconditions	Volunteer performance statistics are generated and available for monitoring and reporting.

3.1.6 Task Service

Description
The Task Service manages operational tasks created during disaster response activities and coordinates volunteer participation.
Functional Requirements

Requirement ID	**FR-TASK-001**
Requirement Name	Task Creation
Description	Organization Coordinators shall create disaster response tasks by providing the task title, description, location, required skills, priority, and deadline.
Actors	Organization Coordinator
Preconditions	Coordinator is authenticated and authorized to create tasks.
Postconditions	A new task is created and stored in the system with the specified details.

Requirement ID	**FR-TASK-002**
Requirement Name	Volunteer Assignment
Description	Organization Coordinators shall assign one or more volunteers to tasks. Assignment decisions shall consider volunteer availability, skills, experience, and location.
Actors	Organization Coordinator
Preconditions	Task exists and eligible volunteers are available.
Postconditions	Selected volunteers are assigned to the task and assignment records are updated.

Requirement ID	**FR-TASK-003**
Requirement Name	Update Task
Description	Organization Coordinators shall modify task details, including the deadline, description, assigned volunteers, and priority. Any changes shall generate notification events through Apache Kafka.
Actors	Organization Coordinator
Preconditions	Task exists and coordinator is authorized to modify it.
Postconditions	Task information is updated and notification events are generated.

Requirement ID	**FR-TASK-004**
Requirement Name	Task Progress
Description	Assigned volunteers shall update task progress. Progress statuses include **Assigned**, **Accepted**, **In Progress**, **Completed**, and **Cancelled**. All progress updates shall include timestamps.
Actors	Volunteer
Preconditions	Volunteer is assigned to the task.
Postconditions	Task progress status and timestamps are updated in the system.

Requirement ID	**FR-TASK-005**
Requirement Name	Task Completion
Description	Completed tasks shall record the completion date, completion notes, assigned volunteers, and resources consumed. The system shall archive completed tasks.
Actors	Volunteer, Organization Coordinator
Preconditions	Task has been completed.
Postconditions	Task is marked as completed, completion details are recorded, and the task is archived.

Requirement ID	**FR-TASK-006**
Requirement Name	Task Monitoring
Description	Organization Coordinators shall monitor pending, active, overdue, and completed tasks. Dashboard statistics shall be updated automatically whenever relevant system data changes.
Actors	Organization Coordinator
Preconditions	Tasks exist in the system.
Postconditions	Task status and dashboard statistics are displayed with up-to-date information.

Requirement ID	**FR-TASK-007**
Requirement Name	Task Reports
Description	The system shall generate reports showing task completion rates, volunteer workload, average completion time, and resource utilization.
Actors	Organization Coordinator, System Administrator
Preconditions	Task and assignment data are available.
Postconditions	Task reports are generated and available for viewing or export.

3.1.7 Notification Service

Description
The Notification Service delivers real-time notifications to users through Apache Kafka events and REST APIs. It keeps users informed about important activities and updates within the system.
Functional Requirements

Requirement ID	**FR-NOT-001**
Requirement Name	Notification Generation
Description	The Notification Service shall generate notifications for user registration, organization approval, resource request updates, resource allocation, volunteer assignment, task completion, and inventory alerts.
Actors	Notification Service
Preconditions	A relevant system event has occurred.
Postconditions	The appropriate notification is generated for the affected users.

Requirement ID	**FR-NOT-002**
Requirement Name	In-App Notifications
Description	Authenticated users shall receive notifications through the web and mobile applications. Each notification shall include a title, description, timestamp, and notification type.
Actors	Authenticated User
Preconditions	User is authenticated and a notification has been generated.
Postconditions	The notification is displayed within the web or mobile application.

Requirement ID	**FR-NOT-003**
Requirement Name	Push Notifications
Description	Mobile users shall receive push notifications for critical events such as emergency requests, task assignments, resource availability, and request approvals.
Actors	Mobile User
Preconditions	User has enabled push notifications and a critical event occurs.
Postconditions	Push notification is delivered to the user's mobile device.

Requirement ID	**FR-NOT-004**
Requirement Name	Email Notifications
Description	The system shall send email notifications for account verification, password reset, organization approval, volunteer approval, and request approval or rejection.
Actors	System, Registered User
Preconditions	A supported event requiring email notification occurs.
Postconditions	Email notification is sent to the user's registered email address.

Requirement ID	**FR-NOT-005**
Requirement Name	Notification Preferences
Description	Users shall configure their notification preferences by selecting one or more notification channels, including email, mobile push, and in-app notifications.
Actors	Registered User
Preconditions	User is authenticated.
Postconditions	Notification preferences are updated and applied to future notifications.

Requirement ID	**FR-NOT-006**
Requirement Name	Notification History
Description	The Notification Service shall maintain notification history. Users shall be able to view notifications, mark them as read, delete notifications, and filter notifications by category.
Actors	Registered User
Preconditions	Notifications exist for the user.
Postconditions	Notification history is updated based on the user's actions.

Requirement ID	**FR-NOT-007**
Requirement Name	Kafka Event Processing
Description	The Notification Service shall subscribe to Kafka topics published by the User Service, Organization Service, Resource Service, Request Service, Volunteer Service, and Task Service. Upon receiving an event, it shall generate and deliver the appropriate notification to affected users.
Actors	Notification Service
Preconditions	A subscribed Kafka event is published by one of the integrated services.
Postconditions	The event is processed successfully and the corresponding notification is delivered to the intended users.

3.2 Usability

The Resource Coordination Platform shall provide a clear, responsive, and easy-to-use interface for users with different levels of technical experience. The platform shall be available through both web and mobile applications while maintaining consistent functionality and appearance. The design shall support quick actions during emergency situations, such as submitting requests, allocating resources, assigning volunteers, and monitoring disaster response activities.

3.2.1 Ease of Use

The system shall be designed so that first-time users can understand the basic functionalities without requiring extensive training.
New users shall be able to:

- Register an account
- Submit a resource request
- View notifications
- Update their profile

within **30 minutes** of initial use.

3.2.2 Consistency

All interfaces shall maintain consistent:

- Navigation menus
- Color schemes
- Typography
- Icons
- Button styles
- Table layouts
- Form structures

This consistency shall reduce the learning curve for users and improve usability.

3.2.3 Responsive User Interface

The system shall provide responsive layouts that automatically adapt to different screen sizes.

Supported platforms include:

Web Platform

- Desktop computers
- Laptop computers
- Tablets

Mobile Platform

- Android smartphones
- Android tablets

The user interface shall remain fully functional across supported devices.

3.2.4 Accessibility

The platform shall comply with common accessibility principles by providing:

- Sufficient color contrast
- Readable font sizes
- Keyboard navigation support for the web application
- Screen reader compatibility where applicable
- Meaningful labels for input controls

3.2.5 Error Prevention

The application shall minimize user errors by:

- Validating user inputs before submission
- Preventing duplicate submissions
- Displaying confirmation dialogs before destructive actions
- Providing informative error messages

Examples include:

- Resource deletion confirmation
- Request cancellation confirmation
- Password validation feedback

3.2.6 Help and Documentation

The platform shall provide:

- User guides
- Frequently Asked Questions (FAQs)
- Tooltips
- Context-sensitive help
- Contact support information

Help resources shall be accessible from every major page.

3.3 Reliability

The Resource Coordination Platform shall operate reliably during disaster response activities, where system availability is important. The architecture shall allow the system to continue operating when individual microservices experience temporary failures.

3.3.1 System Availability

The platform shall achieve a minimum operational availability of 99.5%, excluding scheduled maintenance periods.

3.3.2 Data Integrity

The system shall maintain database consistency throughout all operations.
The system shall prevent:

- Duplicate resource allocations
- Inconsistent request statuses
- Invalid task assignments
- Duplicate organization registrations

Database transactions shall ensure atomicity for critical operations.

3.3.3 Fault Tolerance

The failure of one microservice shall not prevent the remaining services from operating.
For example:

- Failure of the Notification Service shall not prevent request creation.
- Failure of the Volunteer Service shall not affect inventory management.

The API Gateway shall return meaningful error responses when services are unavailable.

3.3.4 Recovery

The platform shall automatically recover from temporary failures whenever possible.
Recovery procedures shall include:

- Automatic service restart
- Retry mechanisms
- Health monitoring
- Error logging

3.3.5 Backup and Restoration

Database backups shall be performed regularly to prevent data loss.
The backup strategy shall support:

- Full backups
- Incremental backups
- Point-in-time recovery

Backup files shall be stored securely.

3.3.6 Audit Logging

Critical system activities shall be logged.
Examples include:

- User login
- Organization approval
- Request approval
- Resource allocation
- Volunteer assignment
- Administrative actions

Logs shall support troubleshooting and auditing.

3.4 Performance and Security

3.4.1 Performance Requirements

3.4.1.1 Response Time

The system shall satisfy the following response time requirements. Table 3.1 summarizes the maximum acceptable response times for common system operations.

Table 3.1: Response Time Requirements

Operation	Maximum Response Time
User Login	2 seconds
User Registration	3 seconds
Submit Resource Request	3 seconds
Update Request Status	2 seconds
Search Resources	3 seconds
Assign Volunteer	3 seconds
Dashboard Loading	5 seconds

3.4.1.2 Concurrent Users

The system shall support at least 500 concurrent users without significant performance degradation.

3.4.1.3 Scalability

The microservice architecture shall allow individual services to scale independently based on workload.
Examples include:

- Scaling the Request Service during disaster situations.

- Scaling the Notification Service during mass alert generation.

3.4.1.4 Database Performance

Frequently queried database columns shall be indexed to improve query performance.
Pagination shall be used for large datasets.

3.4.1.5 Resource Utilization

The system shall efficiently utilize CPU, memory, and network resources to maintain acceptable performance during normal and peak disaster response operations.

3.4.2 Security Requirements

3.4.2.1 Authentication

The system shall authenticate users using an IAM framework that provides centralized authentication and authorization services across all microservices.
The system shall enforce role-based authorization for every protected API endpoint to ensure users can only access resources permitted by their assigned roles.
The IAM framework shall support:

- Secure username and password authentication
- JSON Web Token (JWT) based authentication
- Access token generation and validation
- Refresh token management
- Session management
- Single Sign-On (SSO)
- Secure logout functionality

3.4.2.2 Authorization

Role-Based Access Control (RBAC) shall restrict access to system resources. Permissions shall be assigned according to user roles including:

- Community Member
- Volunteer
- Donor
- Organization Coordinator
- System Administrator

Unauthorized requests shall be rejected.

3.4.2.3 Secure Communication

All communication between clients, the API Gateway, and microservices shall use HTTPS. Sensitive information shall never be transmitted using unsecured protocols.

3.4.2.4 Password Security

Passwords shall:

- Never be stored in plain text
- Follow strong password policies
- Be securely managed by the IAM framework using secure hashing and credential management mechanisms.

3.4.2.5 Data Protection

Sensitive user information shall be protected against unauthorized access. Access to personal data shall comply with organizational privacy policies.

3.4.2.6 API Security

Every protected REST API endpoint shall require:

- Valid JWT
- Role verification
- Token validation

Unauthorized API calls shall return HTTP 401 or HTTP 403 responses.
The API Gateway shall implement rate limiting to reduce the risk of denial-of-service attacks and API abuse.

3.4.2.7 Audit Trail

Security-related events shall be logged including:

- Login attempts
- Failed authentication
- Role modifications
- Administrative operations

3.5 Supportability

The platform shall be designed to make maintenance, deployment, monitoring, and future improvements easier through its modular architecture and standardized development practices.

3.5.1 Maintainability

Each microservice shall be independently maintainable without affecting the remaining services.

3.5.2 Modularity

The system shall separate functionality into independent services including:

- User Service
- Organization Service
- Resource Service
- Request Service
- Volunteer Service
- Task Service
- Notification Service

3.5.3 Logging

Each service shall maintain application logs containing:

- Incoming requests
- Processing time
- Exceptions
- Warnings
- Kafka events

3.5.4 Monitoring

Every microservice shall expose a health-check endpoint. The API Gateway shall periodically verify service availability.

3.5.5 Documentation

REST APIs shall be documented using Open API (Swagger).
API documentation shall include:

- Endpoints
- Request bodies
- Response schemas
- Authentication requirements
- Error codes

3.5.6 Deployment

The system shall support deployment using Docker containers. Individual services shall be deployable independently.

3.6 Design Constraints

The implementation of the Resource Coordination Platform shall comply with technological, architectural, and organizational constraints established during project planning.

3.6.1 Architecture

The implementation of the Resource Coordination Platform shall follow the technological, architectural, and organizational constraints defined during project planning.

3.6.2 Technology Stack

The following technologies shall be used:

Table 3.1: Technology Stack

Component	Technology
Backend	Node.js
Web Framework	Express.js
Frontend	React.js
Mobile Application	Flutter
Database	PostgreSQL (Supabase)
Authentication	IAM Framework (JWT, RBAC, SSO)
API Documentation	Swagger/OpenAPI
Message Broker	Apache Kafka
Containerization	Docker
Version Control	GitHub

3.6.3 Database Constraints

The system shall use a centralized PostgreSQL database hosted on Supabase.
All primary keys shall use UUIDs.
Referential integrity shall be enforced through foreign key constraints.

3.6.4 Communication Constraints

Communication between services shall occur using:

- REST APIs
- JSON message format
- HTTPS
- Apache Kafka for asynchronous messaging

3.6.5 Browser Compatibility

The web application shall support the latest versions of:

- Google Chrome
- Microsoft Edge
- Mozilla Firefox
- Safari

3.6.6 Mobile Compatibility

The mobile application shall support Android devices running Android 10 or later.

3.6.7 Coding Standards

Development shall follow:

- REST API design principles
- Clean Code practices
- Modular programming
- Git branching strategy
- ESLint coding conventions
- Secure software development practices

3.6.8 Single Sign-On (SSO)

The IAM framework shall provide Single Sign-On (SSO) capabilities, allowing authenticated users to

access all authorized microservices without requiring repeated authentication.
The system shall validate the user's authentication token before granting access to protected resources.
Once authenticated, users shall be able to navigate between the web platform, mobile application, and backend services seamlessly until the session expires or the user logs out.

3.7 On-line User Documentation and Help System Requirements

The system shall provide comprehensive online documentation and a context-sensitive help system to assist users in effectively using the platform.
The documentation shall include:

- User Guide
- Frequently Asked Questions (FAQs)
- Tooltips for important interface components
- Error message explanations
- Troubleshooting guidelines
- Contact support information

The help system shall be accessible from all major pages of both the web and mobile applications. Documentation shall be updated when new features or system changes are introduced.

3.8 Purchased Components

The Resource Coordination Platform does not require proprietary purchased software components. The system uses open-source technologies and community-supported frameworks. These technologies help reduce licensing costs while supporting the scalability, maintainability, and long-term operation of the platform.

Table 3.2: Purchased and Open-Source Components

Component	License	Purpose
React.js	MIT	Web UI
Flutter	BSD	Mobile App
PostgreSQL	PostgreSQL License	Database
Apache Kafka	Apache License 2.0	Event Streaming
Docker	Apache License 2.0	Containerization

All listed components shall be maintained using stable releases and updated regularly to address security vulnerabilities and performance improvements.

3.9 Interfaces

All interfaces shall provide clear navigation, consistent layouts, validation messages, confirmation dialogs, and informative error messages to make the system easier to use and reduce user errors.

3.9.1 User Interfaces

The Resource Coordination Platform shall provide responsive and easy-to-use interfaces through both web and mobile applications. The interfaces shall support different user roles while maintaining a consistent appearance and navigation structure.
The user interface shall provide:

- Login and user registration screens
- Role-based dashboards
- Profile management pages
- Organization management interfaces
- Resource request submission forms
- Donation registration interfaces
- Resource inventory management pages
- Volunteer registration and profile management
- Task management dashboards

- Notification center
- Reports and analytics dashboards

The interfaces shall also provide:

- Responsive layouts for desktop, tablet, and mobile devices
- Form validation and informative error messages
- Confirmation dialogs before destructive actions
- Search, filtering, sorting, and pagination
- Consistent navigation menus and icons
- Role-based menu visibility
- Accessibility support for keyboard navigation and readable fonts

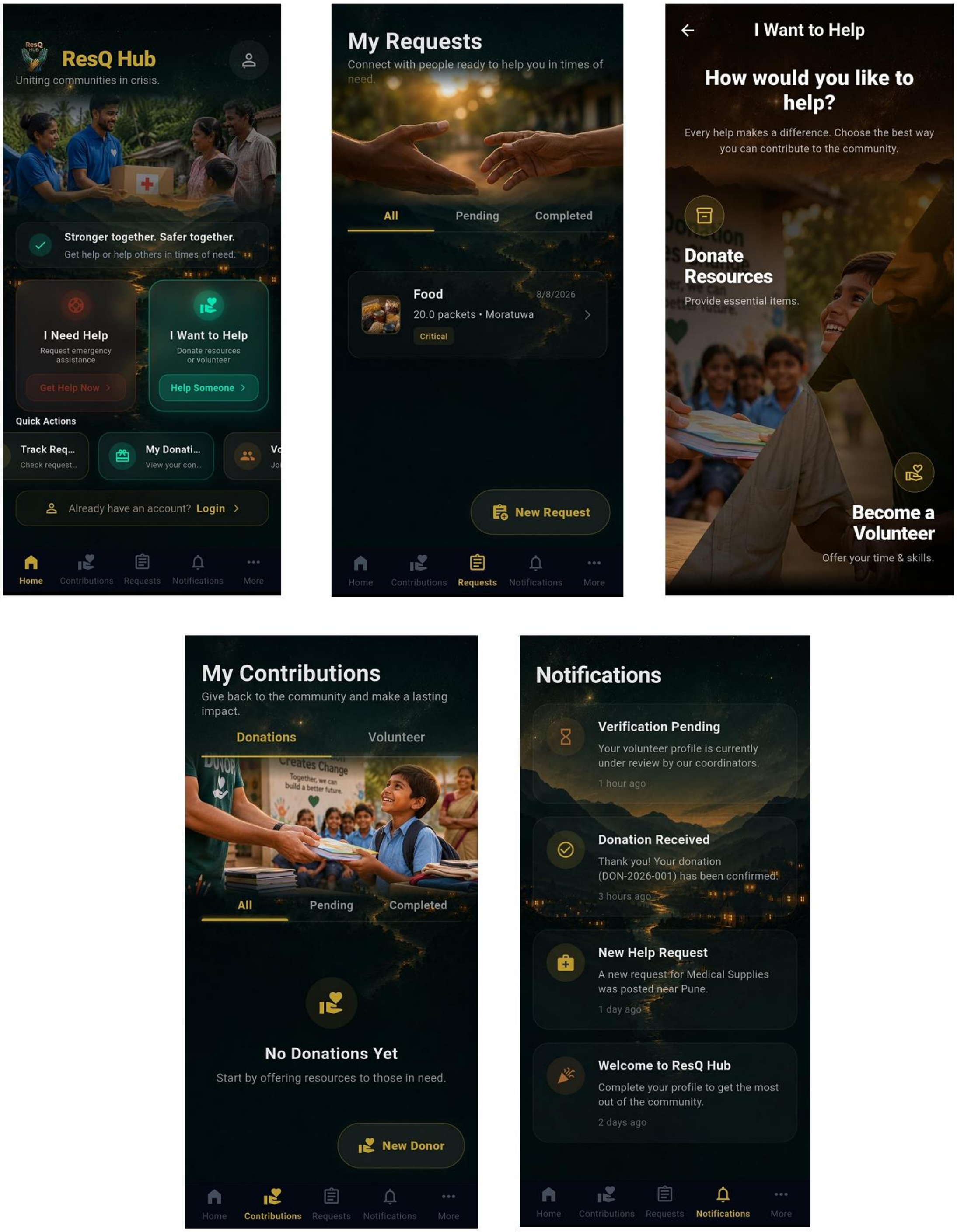

Figure 3.1: Sample User Interface of Mobile Application

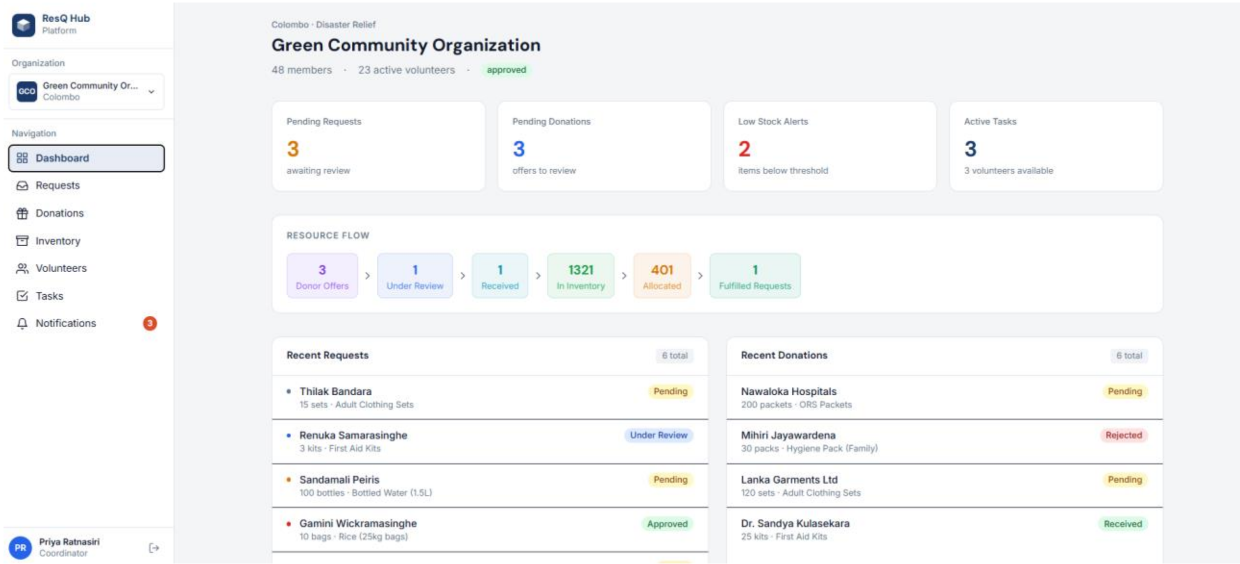

Figure 3.2: Organization Coordinator's dashboard

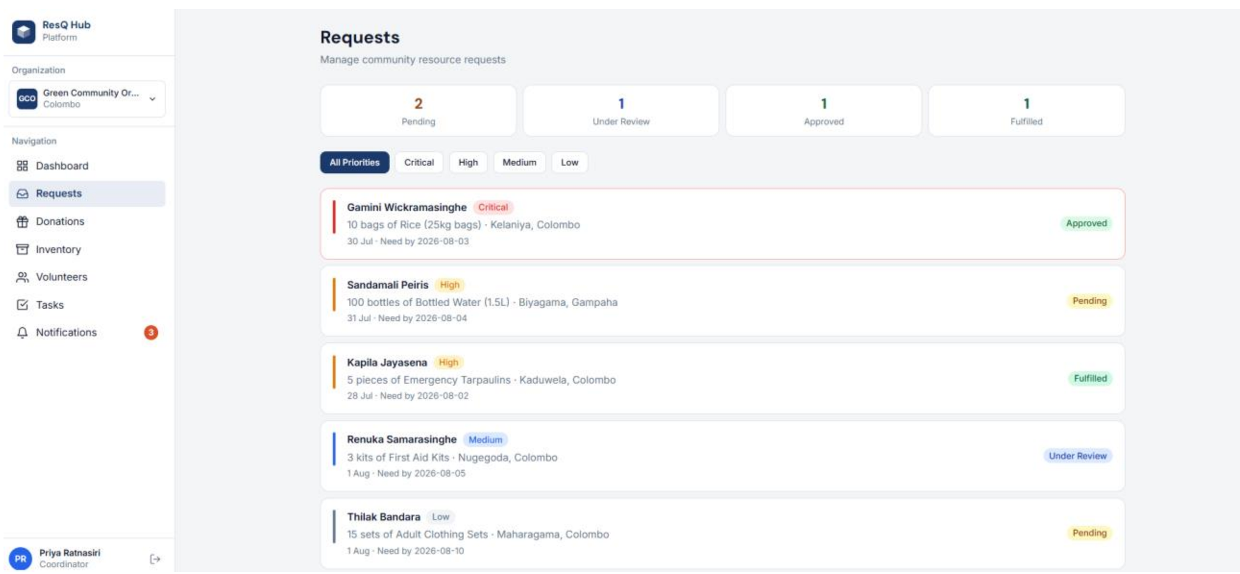

Figure 3.3: Organization Coordinator's request view

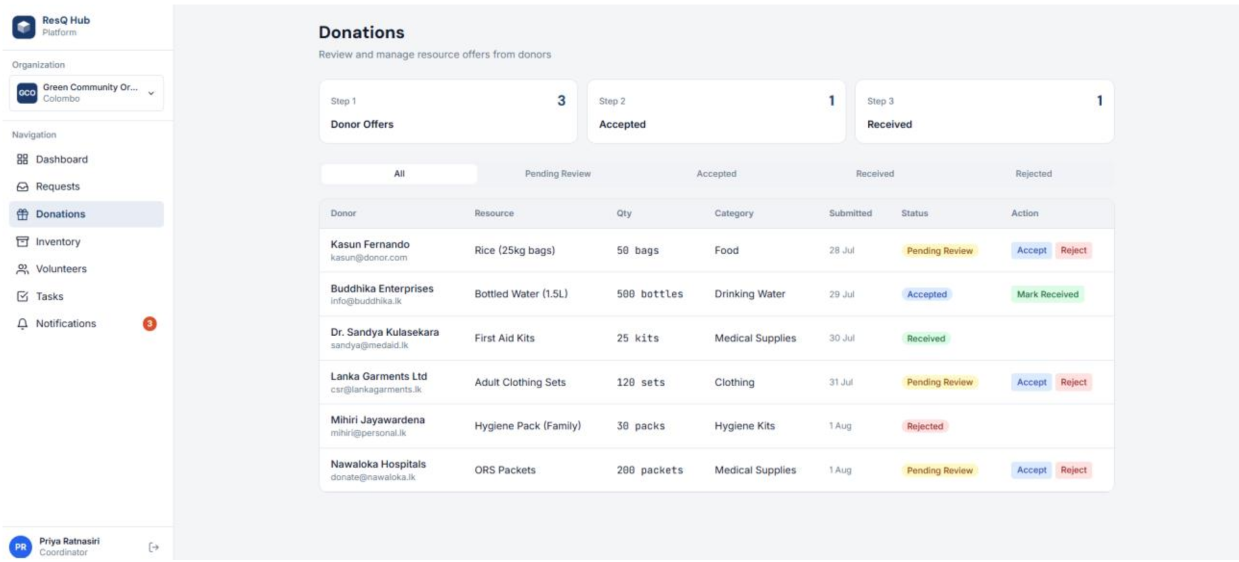

Figure 3.4: Organization Coordinator's donation view

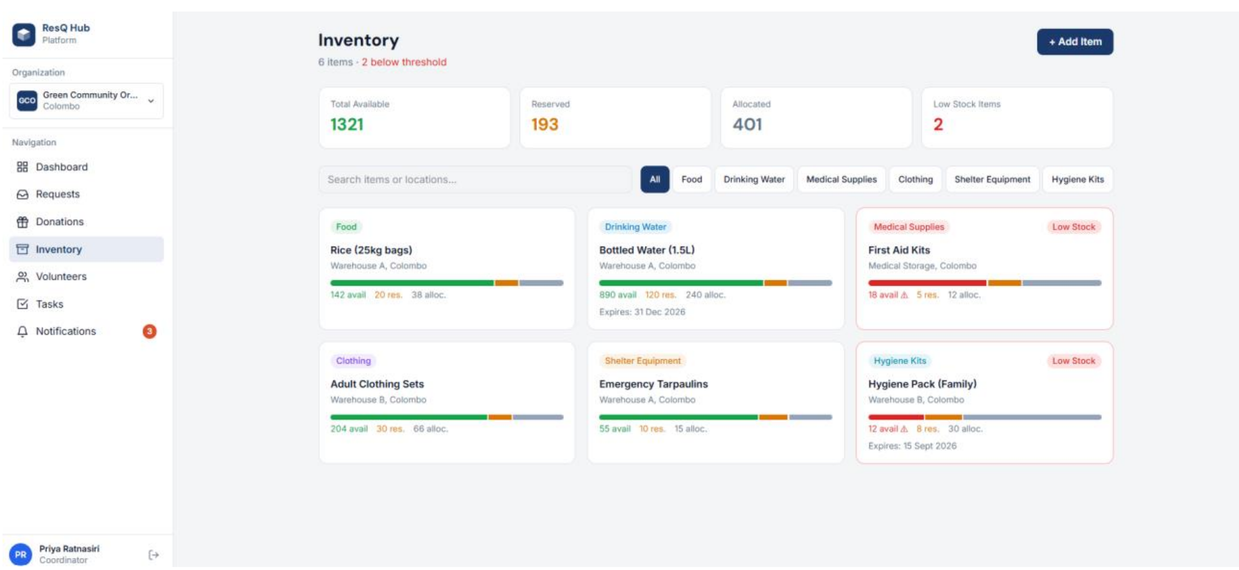

Figure 3.5: Organization Coordinator's inventory view

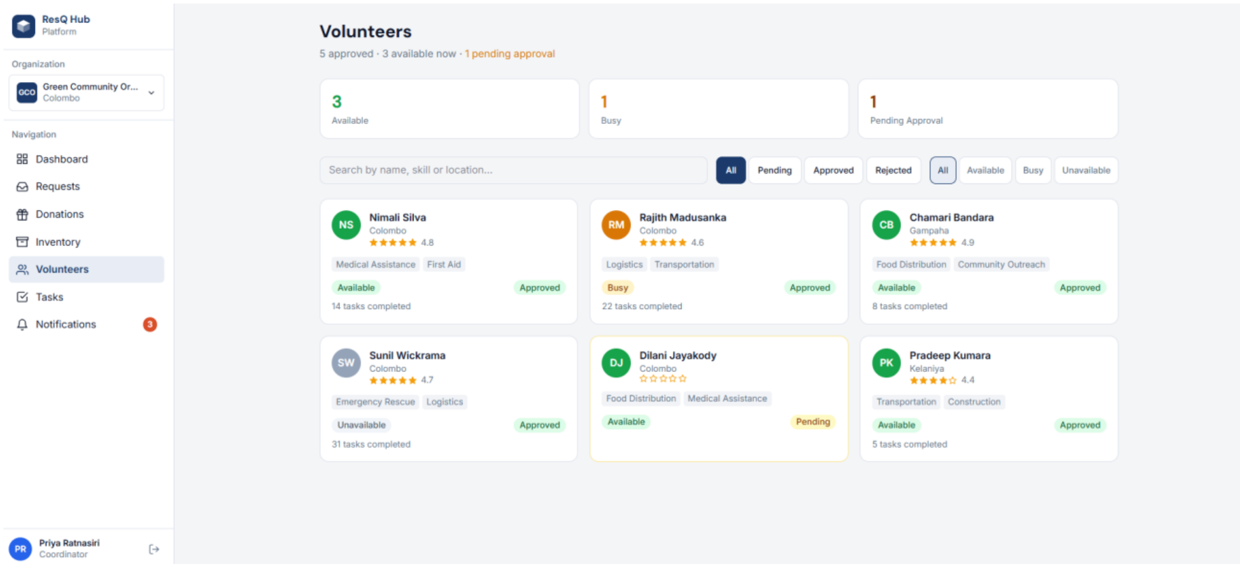

Figure 3.6: Volunteer management view

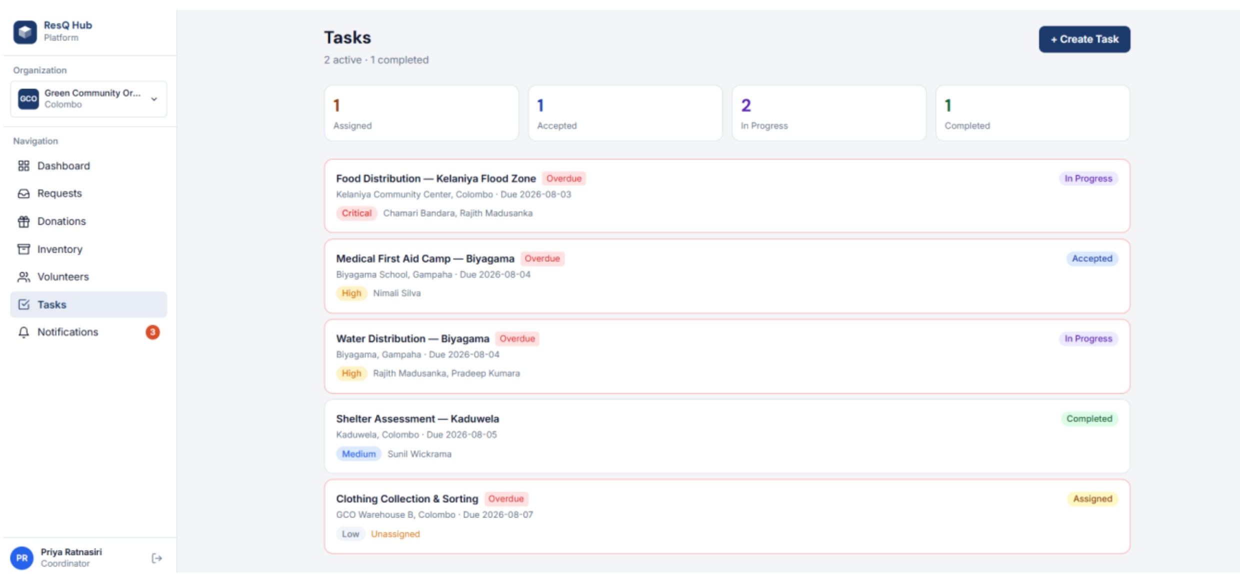

Figure 3.7: Task Creation and view current task view

3.9.2 Hardware Interfaces

The system shall not require specialized hardware. Users shall access the platform using desktop computers, laptops, tablets, and Android mobile devices. Minimum hardware requirements include:

- Dual-Core Processor or higher
- 4 GB RAM
- Stable Internet connection

- Minimum screen resolution of 1366 × 768 for the web application
- Android 10 or later for the mobile application

3.9.3 Software Interfaces

The system shall communicate with the following software components:

- React.js web application
- Flutter mobile application
- API Gateway
- Backend microservices
- IAM framework
- PostgreSQL/Supabase database
- Apache Kafka
- Open API/Swagger documentation

The system shall use REST APIs for communication between clients and backend services where applicable.

3.9.4 Communications Interfaces

The system shall use HTTPS for secure client-server communication. REST APIs shall use JSON for data exchange. Apache Kafka shall be used for asynchronous communication between microservices where asynchronous event processing is required. REST APIs shall exchange data using JSON over HTTPS. JWT-based authentication provided by the IAM framework shall be used to secure protected API requests.

3.10 Database Requirements

The system shall use PostgreSQL as the relational database management system, hosted through Supabase. The database shall store information related to users, organizations, roles, permissions, help requests, request status history, donations, inventory, inventory transactions, volunteers, volunteer tasks, resource allocations, notifications, and audit logs.
The database shall maintain data integrity using primary keys, foreign keys, unique constraints, validation rules, and referential integrity.
The database shall support ACID-compliant transactions for critical operations such as resource allocation, request approval, and inventory updates.
Database access shall be restricted to authorized system components and users according to their roles and permissions. The database shall support backup and recovery mechanisms to protect against data loss.

3.11 Licensing, Legal, Copyright, and Other Notices

The project shall comply with the applicable licenses and usage conditions of all third-party software, frameworks, libraries, and services used during development. The project team shall not use copyrighted software, content, or other third-party materials without appropriate permission or licensing. User information and system data shall be protected against unauthorized access and disclosure. Third-party logos, trademarks, and copyrighted materials shall be used according to their applicable terms and conditions.

3.12 Applicable Standards

The project shall follow applicable software engineering and technical standards, including:

- IEEE Std 29148-2018 – Software Requirements Specification
- REST principles for API design.
- Open API specification for REST API documentation.
- HTTPS for secure communication.
- JSON (RFC 8259)
- RFC 7519 – JSON Web Token (JWT) for token-based authentication where applicable.
- PostgreSQL standards and conventions for relational database implementation.
- OWASP Secure Coding Guidelines

4. Supporting Information

This chapter provides additional information that helps explain and use the requirements specified in this SRS.

The supporting documents and diagrams for the system may include:

- Use Case Diagram
- Use Case Specifications
- System Architecture Diagram
- Entity Relationship Diagram (ERD)
- Database Schema
- API Documentation
- User Interface Designs
- Requirement Traceability Matrix

These supporting artefacts provide additional details about the system requirements and complement the requirements described in this document.